FitBuddy – Project Description

Understand the Project Requirements

FitBuddy's AI model needed to:

- Generate structured 7-day workout plans
- Provide concise and actionable nutrition tips
- Update workout plans based on user feedback
- Perform well in real-time, web-based interaction

Model Evaluation

We initially experimented with several Hugging Face models:

Model	Purpose
T5 & BART	Text generation (workout/nutrition)
DistilBERT	Potential intent classification
BLOOM	General purpose generation

While these models showed promise in basic text generation, they had notable limitations.

These models required:

- Hosting with Hugging Face or local GPU setup
- Fine-tuning or complex prompting to achieve structured plans

Final Model Selection

After comparative testing, we selected Google Gemini models for production:

Gemini 1.5 Pro

- Used for structured 7-day workout generation and plan updating
- Delivers rich, context-aware, formatted responses reliably

Gemini Flash

- Used for generating fast, practical nutrition tips
- Lightweight and efficient for quick API calls

Reasons for Choosing Gemini

- API-based, easy to integrate with FastAPI
- Naturally understands structured formatting (days, sections)
- High accuracy with minimal prompt tuning

- Scalable and fast for feedback-based revisions